Conditional Probability

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Motivation

- how does the probability of an event change if we have extra information?

Example

- Toss a fair coin 3 times. Suppose we are told that the first toss was heads. Given this information how should we compute the probability of 3 heads?

answer: We have a new (reduced) sample space: $\Omega' = \{HHH, HHT, HTH, HTT\}$. All outcomes are equally likely, so

$$P(3 \text{ heads given that the first toss is heads}) = 1/4.$$

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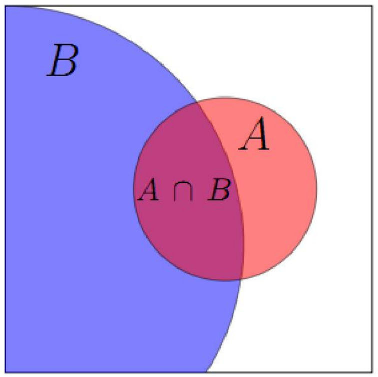
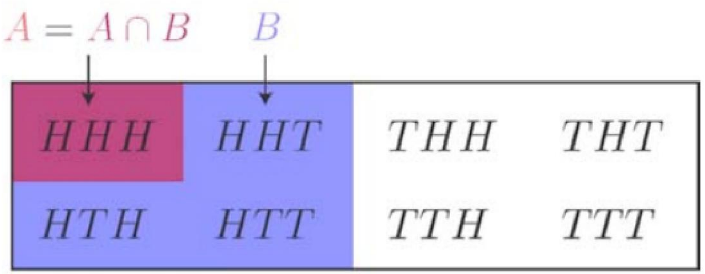




Formal definition of conditional probability

- Let A and B be events. We define the conditional probability of A given B as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) \neq 0$$

"the conditional probability of A given B" or
"the probability of A conditioned on B"

Example

- Draw two cards from a deck. Define the events:

  S1="first card is a spade (黑桃)" ,
  S2="second card is a spade".

  What is P(S2|S1)?

Multiplication (Chain) Rule

$$P(A \cap B) = P(B \mid A) \cdot P(A)$$

$$\mathrm{P}(A_n \cap \ldots \cap A_1) = \prod_{k=1}^{n} \mathrm{P}\left(A_k \,\middle|\, \bigcap_{j=1}^{k-1} A_j\right)$$

Example

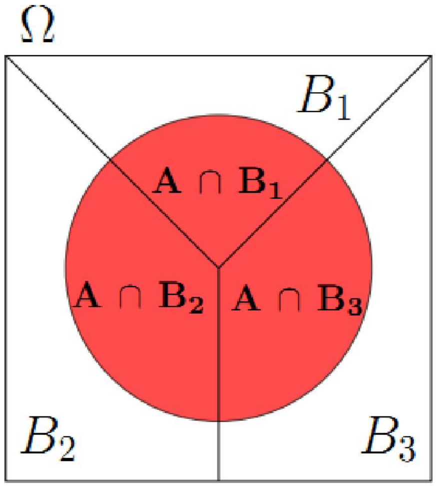
- Draw two cards from a deck. Define the events: S1="first card is a spade" , S2="second card is a spade". What is $P(S1 \cap S2)$?

Law of Total Probability

- Suppose the sample space Ω is divided into 3 disjoint events B1, B2, B3. Then for any event A:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3)$$

$$P(A) = P(A|B_1)\,P(B_1) + P(A|B_2)\,P(B_2) + P(A|B_3)\,P(B_3)$$



Remark

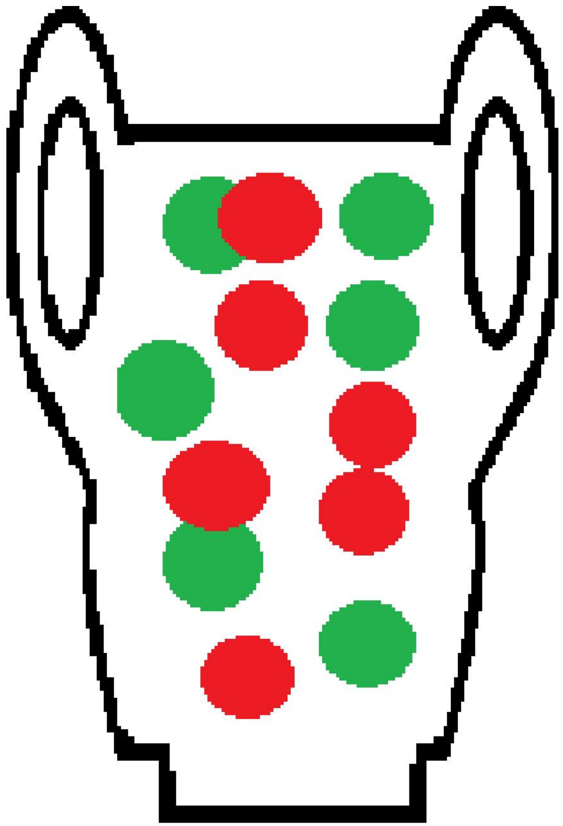
- The law of total probability holds if we divide Ω into any number of events, so long as they are disjoint and cover all of Ω. Such a division is often called a partition of Ω.

Example

- An urn contains 5 red balls and 2 green balls. Two balls are drawn one after the other. What is the probability that the second ball is red?

Pólya Urns Model

In an urn model, objects of real interest (such as atoms, people, cars, etc.) are represented as colored balls in an urn or other container.
One ball is drawn randomly from the urn and its color observed; it is then returned in the urn, and additional balls are added to the urn based on certain strategy, and the selection process is repeated. Questions of interest are the evolution of the urn population and the sequence of colors of the balls drawn out.



Tree Method

- An urn contains 5 red balls and 2 green balls. A ball is drawn. If it's green, a red ball is added to the urn and if it's red, a green ball is added to the urn (The original ball is not returned to the urn.) Then a second ball is drawn. What is the probability the second ball is red?

answer: The law of total probability says that $P(R_2)$ can be computed using the expression in Equation (4). Only the values for the probabilities will change. We have

$$P(R_2|R_1) = 4/7, \qquad P(R_2|G_1) = 6/7.$$

Therefore,

$$P(R_2) = P(R_2|R_1)P(R_1) + P(R_2|G_1)P(G_1) = \frac{4}{7} \cdot \frac{5}{7} + \frac{6}{7} \cdot \frac{2}{7} = \frac{32}{49}.$$

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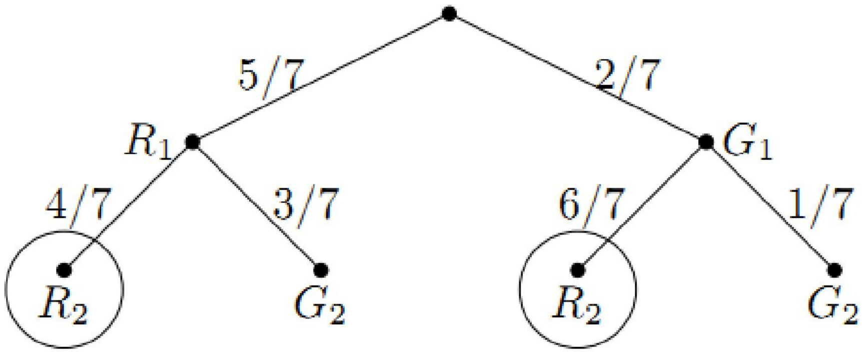


Independence

- Two events are independent if knowledge that one occurred does not change the probability that the other occurred. Informally, events are independent if they do not influence one another.

$$A \text{ is independent of } B \quad \text{if} \quad P(A|B) = P(A)$$

Formal definition of independence

- Two events A and B are independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

Example

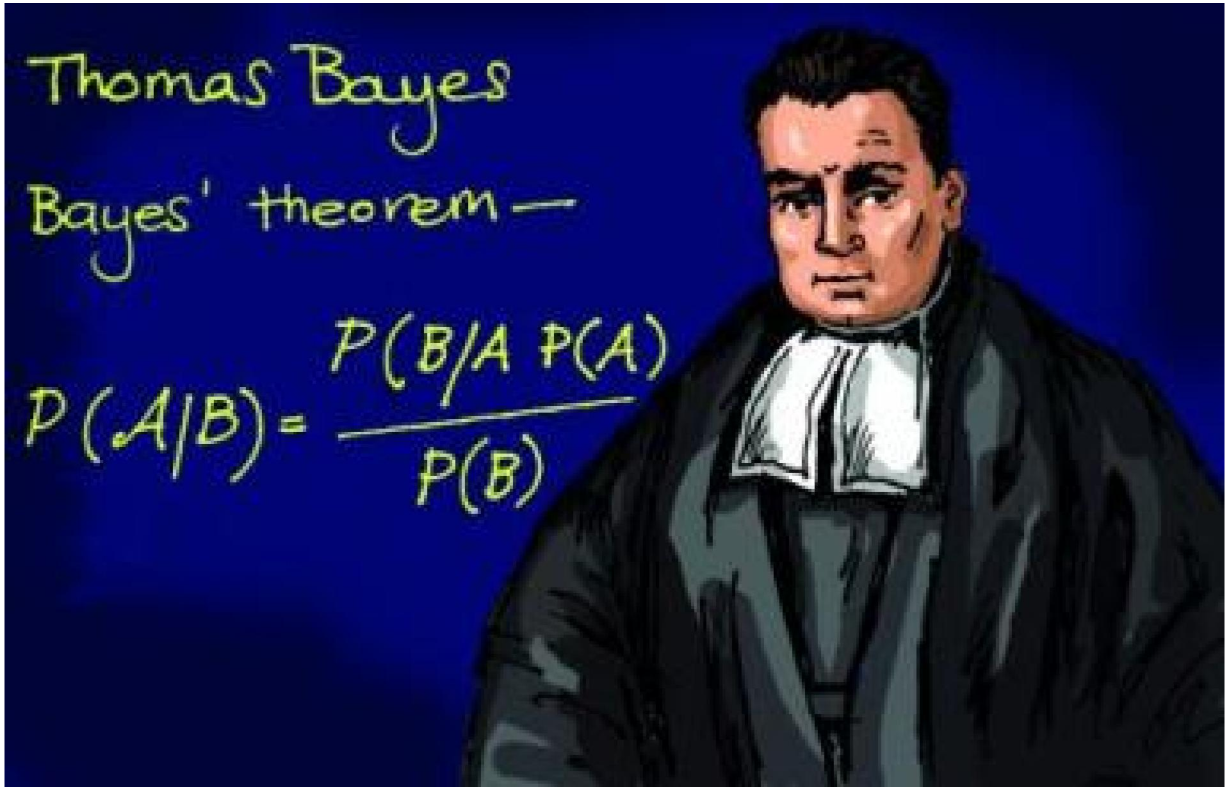
- Toss a fair coin 3 times. Let H1= 'heads on first toss' and A = 'two heads total'. Are H1 and A independent?

answer: We know that $P(A) = 3/8$. Since this is not 0 we can check if the formula in Equation 5 holds. Now, $H_1 = \{\text{HHH, HHT, HTH, HTT}\}$ contains exactly two outcomes (HHT, HTH) from A, so we have $P(A|H_1) = 2/4$. Since $P(A|H_1) \neq P(A)$ these events are not independent.

Remark

- Independence for more than two events

Bayes' Theorem



Bayes' Theorem

- For two events A and B, Bayes' theorem (also called Bayes' rule and Bayes' formula) says

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

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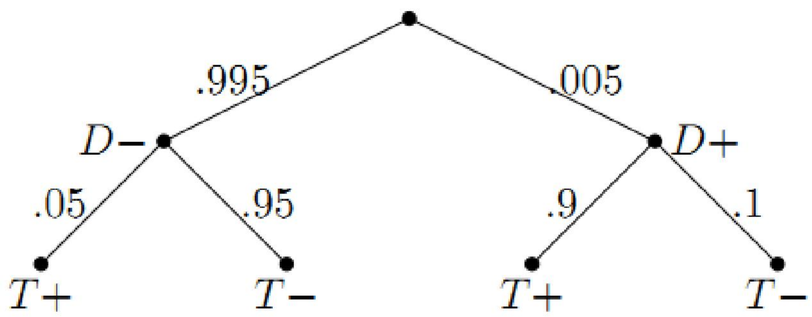
Remarks

- Bayes' rule tells us how to 'invert' conditional probabilities, i.e. to find P(B|A) from P(A|B).
- In practice, P(A) is often computed using the law of total probability.

The Base Rate Fallacy(基本比率谬误)

- Consider a routine screening test for a disease. Suppose the frequency of the disease in the population (base rate) is 0.5%. The test is highly accurate with a 5% false positive rate and a 10% false negative rate. You take the test and it comes back positive. What is the probability that you have the disease?



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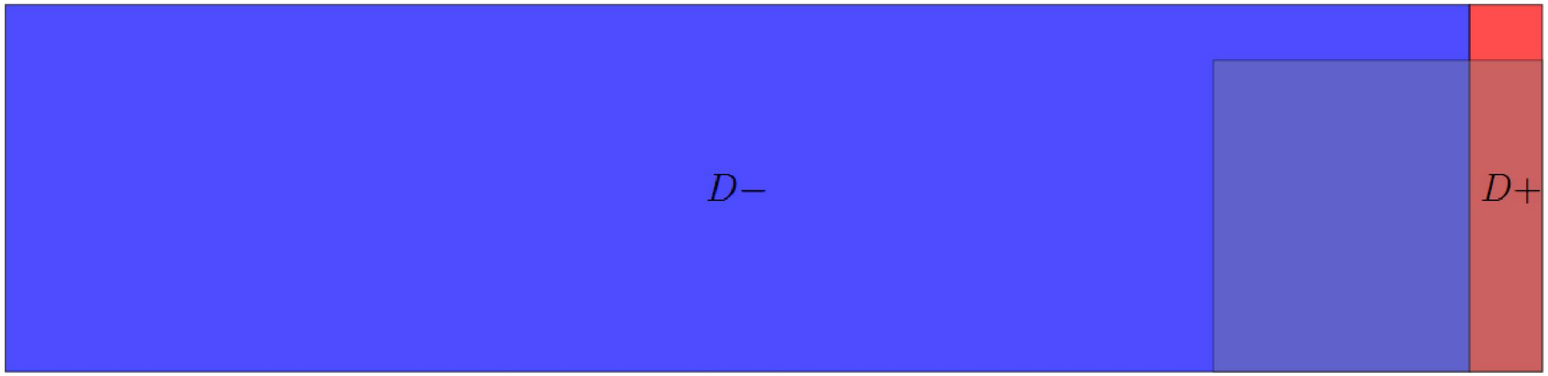

The shaded rectangle represents the the people who test positive